The Backtest That Lied:

A Forensic Investigation of Leakage, Implementation Failure, and Liquidity Dependence in A-Share Machine Learning

BRS Student Researcher

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1 Abstract

A near-perfect cross-sectional prediction signal—monthly rank IC 0.940 across 96 consecutive out-of-sample months—collapsed to 0.100 after four detected data leakages were corrected. The surviving signal, a mean out-of-sample IC of 0.108 with Newey–West $t = 10.5$, represented genuine low-liquidity stock-selection ability, concentrated overwhelmingly in the bottom quintile by trading volume (+48.1 of all holdings). However, when translated into executable next-open portfolios with realistic transaction costs, the strategy produced negative absolute returns in the liquid U50 universe (net annual -0.9, maximum drawdown +61.3). Residualization marginally improved liquid-universe performance, and multi-horizon targets failed to rescue scalability. The OHLCV-only hypothesis was formally terminated as a core strategy. The primary contribution is methodological: a demonstrated system for detecting self-deception in quantitative research, and the discipline to reject a favored hypothesis when evidence turns against it.

2 Introduction

A model can predict returns and still fail as a strategy.

This project began with a straightforward question: can daily price and volume data—commonly called OHLCV—predict next-month A-share stock returns well enough to build a profitable trading strategy?

The initial answer appeared to be an unambiguous yes. A gradient-boosted tree model trained on 20 technical indicators produced a rank information coefficient of 0.940 across 96 consecutive out-of-sample months. Every single month was positive. The probability of this occurring by chance is approximately $2^{-96} \approx 10^{-29}$.

But the result was wrong.

Over six experimental phases spanning 200+ model fits, four distinct data leakages were discovered and corrected: contemporaneous target timing, own-stock factor contamination, survivorship bias from snapshot-based universe construction, and temporal encoding in raw feature levels. After all corrections, the apparent near-perfect predictive power collapsed by 89

What followed was not a story of failure, but of methodical investigation. A frozen research protocol was established: walk-forward validation on three independent folds, 20-seed ensemble averaging, cross-sectional rank normalization, and pre-registered decision gates that forbade post-hoc optimization. An adversarial null infrastructure was constructed—iid Gaussian noise tests, constant-label checks, and permutation audits—to ensure any remaining signal was genuine.

The result of this investigation was a conclusion more nuanced than either "it works" or "it doesn't work."

(1) **Cross-sectional predictability existed.** The retained IC of 0.108 across 96 pooled out-of-sample months was statistically robust under Newey–West adjustment ($t = 10.5$) and block-bootstrap inference.

(2) **The signal was concentrated in illiquid stocks.** ADV20 quintile decomposition revealed that +48.1 of holdings resided in the lowest-liquidity quintile (median daily volume ¥19M). Within this quintile, genuine stock selection was demonstrated (+2.7 monthly excess).

(3) **Liquid universes remained unprofitable.** Executable next-open portfolios in the top 50% and top 30% by ADV20 produced negative absolute returns despite positive benchmark-relative excess.

(4) **Longer horizons did not rescue scalability.** Three-month and six-month prediction targets, tested through the identical execution engine, produced negative liquid-universe returns (H3: -21.7, H6: -15.2 net annual).

On the basis of this evidence, the OHLCV-only core strategy was terminated. The branch was closed not because the model predicted nothing—it predicted something real—but because that something could not be converted into a scalable investable strategy under the constraints of real trading.

This report documents the investigation, the evidence architecture, and the decision framework that led to this conclusion. It is offered not as a trading strategy, but as a case study in research methodology: how to build systems that can prove you wrong, and why knowing when to stop is itself a skill.

3 Research Question

The primary research question is:

> *Can daily OHLCV data produce a statistically valid, economically executable, and scalable A-share trading strategy?*

This question decomposes into three hierarchical tests:

T1: Is the predictive signal real? — Can cross-sectional return rankings be predicted out-of-sample with statistical significance, after correcting for known data leakages (survivorship bias, look-ahead bias, own-stock contamination, calendar encoding)?

T2: Can the signal survive execution and costs? — When translated into an executable portfolio with next-open timing, realistic buy/sell costs, and a 90/80 no-trade band, does the signal produce positive absolute returns?

T3: Can the signal survive liquidity and capacity constraints? — Does the strategy work in tradable universes (top 50% and top 30% by ADV20), or is performance dependent on stocks too illiquid to trade at meaningful scale?

Each test was evaluated against pre-registered gates. No gate was changed after seeing results. Experiments that failed a gate were classified as rejected; the hypothesis branch was terminated when all rescue attempts (residualization, multi-horizon targeting) failed the same gates.

4 Data and Methodology

4.1 Data

The research used the RQAlpha A-share daily market-data bundle, covering 5,511 historically listed stocks. A point-in-time (PIT) universe was constructed without survivorship filters, resulting in 4,073 stocks appearing across the full sample period. Daily OHLCV data was aggregated to monthly frequency.

Important: True point-in-time fundamental data (market capitalization, book value, earnings, industry classification) was unavailable. All analysis was conducted using price and volume data only. This limitation is fundamental to the project's conclusion.

4.2 Feature Set

Twenty ranked technical indicators were computed at monthly frequency:

- **Momentum:** 5-, 10-, 20-, 60-, and 120-day returns (M5, M10, M20, M60, M120)
- **Volatility:** 5-, 20-, and 60-day standard deviation of returns (S5, S20, S60)
- **Volume:** 5- and 20-day average turnover ratios (V5, V20)
- **Moving Average:** 20- and 60-day MA deviation (MA20, MA60)
- **Channel Position:** Position within 20-day high-low range (POS)
- **RSI:** 14-day relative strength index (RSI)
- **Turnover:** Current and 5-day average turnover (TN, TN5)
- **Reversal:** 1- and 5-day reversal (REV1, REV5)
- **Cross-sectional:** 20-day cross-sectional rank (X20)
- **Skewness:** 20-day return skewness (SK20)
- **Breadth:** Number of advancing stocks over 20 days (N20)

All features underwent cross-sectional rank normalization to $[-1, 1]$ within each month. This normalization was recomputed separately for each validation month during training to prevent information leakage.

4.3 Model

LightGBM gradient-boosted trees were used with the following frozen hyperparameters:

Parameter	**Value**
Number of estimators	200
Learning rate	0.02
Maximum depth	3
Number of leaves	8
Subsample fraction	0.7
Column subsample	0.7
L1 regularization (α)	1.0
L2 regularization (λ)	10.0

A 20-seed ensemble (seeds 42–241) was used for each fold. No hyperparameter tuning was performed; the parameters were frozen before any results were observed.

4.4 Validation Protocol

Three temporally ordered walk-forward folds were used:

Fold	Training	Validation
Fold 1	2010–2016	2017–2019
Fold 2	2013–2019	2020–2021
Fold 3	2015–2021	2022–2024

4.5 Inference

- **Information Coefficient (IC):** Cross-sectional Spearman rank correlation between predicted and realized returns, computed monthly.
- **ICIR:** Mean IC divided by standard deviation of IC.
- **Newey–West:** Heteroskedasticity- and autocorrelation-consistent standard errors with 3 and 6 lags.
- **Block Bootstrap:** 1,000 resamples with block length of 6 months, 95% confidence intervals.

4.6 Null Infrastructure

Three adversarial tests were applied to every specification:

(1) **iid Gaussian null:** Replace target returns with independent standard normal draws. Expected: IC ≈ 0.

(2) **Constant-label test:** Assign identical labels to all observations. Expected: standard deviation of predictions ≈ 0.

(3) **Permutation audit:** Shuffle labels across stocks within each month, retrain, check correlation with original predictions. Expected: ≈ 0.

4.7 Portfolio Construction

Monthly rebalancing with 90/80 no-trade bands:
- At month-end: rank stocks by ensemble prediction.
- Select top 10% by rank percentile.
- Enter at $\geq$ 90th percentile; retain at $\geq$ 80th percentile.
- Execute at next trading-day open.
- Equal weight within portfolio.

4.8 Cost Scenarios

Scenario	Buy (bp)	Sell (bp)
Gross	0	0
Base	13	18
Stress	22.5	27.5

Cost components: commission 0.03%, stamp tax 0.05%, slippage 0.1%.

4.9 Evidence Architecture

All numerical claims in this report are generated from `data/evidence.json`, a machine-readable evidence layer containing 17 metrics, 3 collapse records, and full provenance information (experiment IDs, artifact paths, validation statuses). The `scripts/validate_evidence.py` validator and `tests/test_evidence.py` unit tests enforce structural integrity and reference resolution.

5 The Three Collapses

This section documents the three stages through which an apparently extraordinary result was progressively dismantled.

5.1 Collapse I: Statistical Illusion

Initial belief: A LightGBM model with 20 technical features could predict next-month cross-sectional A-share returns with a monthly rank IC of 0.940.

What was wrong: The model was predicting the contemporaneous return, not the forward return. The target variable was misaligned: features at the end of month t were being used to predict the return *during* month t, not month $t+1$.

Detection: A date-shift scan was performed, computing the IC at shift k for $k \in \{-3, \ldots, +3\}$. The IC peaked at $k = 0$ (contemporaneous) rather than $k = +1$ (forward).

Correction: The target variable was shifted forward by one period. After correction, the IC collapsed from 0.940 to 0.100.

Interpretation: 89% of the apparent predictive power was a mechanical illusion created by predicting returns that had already occurred. The absolute decline was 0.84 IC points. The remaining signal, while real, was an order of magnitude smaller than the original result suggested.

5.2 Collapse II: Economic Implementation

Initial belief: A statistically significant IC of 0.108 should translate into positive portfolio returns.

Results: When implemented as an executable next-open portfolio with realistic costs (31bp round-trip), the strategy in the top-50%-by-ADV20 universe (U50) produced a net annual return of -0.9 with a maximum drawdown of +61.3. The benchmark (equal-weight U50) returned -5.9 annually.

Interpretation: Positive excess over a declining benchmark is not sufficient. The absolute return matters for an investor who can choose between the strategy and a risk-free alternative.

5.3 Collapse III: Scalability

Initial belief: The strategy was profitable in the full universe (U100 net annual +6.8) and simply needed the right execution universe.

What was wrong: The full-universe performance was concentrated overwhelmingly in the lowest-liquidity quintile. ADV20 quintile decomposition revealed that +48.1 of all holdings were in Q1 (median daily turnover ¥19M). Within Q1, genuine stock selection was demonstrated (+2.7 monthly excess over Q1 equal-weight). However, in the liquid U50 universe, returns turned negative.

Interpretation: The model had learned to identify stocks that would perform well—but only among the lowest-liquidity names. The strategy could not be scaled because the alpha existed only where capacity was severely constrained.

6 Portfolio Implementation

6.1 Portfolio Results

Table **??** presents the primary portfolio outcomes under the base cost scenario (31bp round-trip).

The U100 portfolio achieved positive net returns but with extreme concentration in low-liquidity stocks. The U50 portfolio failed to produce positive absolute returns, despite generating positive benchmark-relative excess.

Table 1: Portfolio performance under base cost scenario (31bp round-trip).

Portfolio	Net Annual	Sharpe	Max DD	Excess
U100 RAW (base31)	+6.8	+0.31	+46.3	+6.9
U50 RESID (base31)	-0.9	-0.04	+61.3	+4.9
U50 EW benchmark	-5.9	-0.26	—	—

6.2 The Excess-Return Trap

U50 RESID generates +4.9 annual excess over a declining benchmark of -5.9. Positive excess over a poor benchmark is not evidence of investment skill.

6.3 Cost Analysis

Transaction costs accounted for approximately 4.2 percentage points of annual return drag. Primary drivers: stamp tax (5bp on sells, 3.2% p.a.) and commission plus slippage (1.0% p.a.).

7 Liquidity Attribution

7.1 ADV20 Quintile Decomposition

To determine where the model's predictive ability was concentrated, all predicted stocks were assigned to quintiles based on their 20-day average daily trading volume (ADV20).

Table 2: Liquidity quintile composition of the U100 portfolio.

Quintile	Median ADV (M¥)	Holdings Share	Net Annual
Q1 (lowest ADV)	19	+48.1	+10.4%
Q2	41	25.8%	+3.9%
Q3	75	14.0%	-0.6%
Q4	145	7.8%	-3.5%
Q5 (highest ADV)	386	4.2%	-2.6%

7.2 Within-Quintile Selection

The model demonstrated genuine stock-selection ability within the lowest-liquidity quintile: Q1 holdings outperformed the Q1 equal-weight benchmark by +2.7 per month.

The within-quintile excess decays monotonically across quintiles: Q2 excess was negative (-2.8% monthly).

7.3 Interpretation

The signal behaves as a low-liquidity stock selector. This is consistent with Leippold, Wang, and Zhou (2022), who document that short-horizon machine-learning predictability in Chinese equities is strongest among the lowest-liquidity stocks. The ADV20-based finding should be interpreted as a liquidity effect, not a size-factor finding (true point-in-time market capitalization was unavailable).

7.4 Scalability Barrier

The concentration in Q1 creates a fundamental capacity constraint. With median daily turnover of ¥19M per stock, the strategy could deploy approximately ¥5–10M before market impact became material.

8 Residual and Horizon Tests

8.1 Cross-Fitted Residual Target (RESID)

The RESID experiment followed Gu, Kelly, and Xiu (2020): instead of predicting raw forward returns, the model was trained to predict the residual from a cross-sectional regression on four Chinese equity factors.

Results: Residualization produced a modest improvement in the liquid U50 universe. However, the improvement was insufficient to cross the profitability gate: U50 RESID remained negative in absolute terms at -0.9, and maximum drawdown remained at +61.3—above the 25% threshold.

Conclusion: Residualization is retained as a cleaner research baseline but rejected as a strategy fix.

8.2 Multi-Horizon Targets (H3 and H6)

The multi-horizon experiment tested whether training on 3-month and 6-month cumulative returns could shift the signal into more liquid stocks.

Results: Neither horizon produced positive U50 net returns. H3 U50 returned -21.7 annually; H6 U50 returned -15.2 annually. Turnover was lower at longer horizons but the gross signal was insufficient.

Conclusion: Multi-horizon targeting does not rescue OHLCV-only scalability.

9 The Signal Was Real. The Strategy Was Not.

Table 3: Decision gates and outcomes.

Criterion	Result	Gate
Predictive ranking	Passed	IC > 0, NW $t > 2$
Time-aware significance	Passed	Bootstrap CI excludes 0
Within-Q1 selection	Passed	Excess > 0
Positive U50 return	Failed	Net < 0
Acceptable drawdown	Failed	DD $> 25\%$
Scalable capacity	Failed	Q1 holds +48.1
Decision	**TERMINATED**	OHLCV branch closed

The frozen baseline retained a mean out-of-sample IC of 0.108 with Newey–West $t = 10.5$.

9.1 Prediction vs. Decision

Each implementation step revealed constraints: statistical significance survived corrections; execution introduced timing gaps; costs consumed 4+ p.p. of annual return; liquidity concentrated the signal in stocks where capacity was severely constrained; drawdown remained above threshold.

9.2 The Discipline of Stopping

The pre-registered decision gates prevented post-hoc optimization. When all rescue attempts failed the same gates, termination was the only honest decision.

10 Limitations

The following limitations apply to all claims in this report:

(1) **Data scope:** Only daily OHLCV data was available. True point-in-time fundamental data (market capitalization, book value, earnings, accruals, industry classification, analyst forecasts) was not accessible during the research period. The conclusion that OHLCV-only cannot produce a scalable strategy does not extend to strategies incorporating economically independent data sources.

(2) **Unavailable market capitalization:** The ADV20-based liquidity findings should not be interpreted as size-effect findings. Without point-in-time market capitalization, it is not possible to distinguish between a liquidity effect and a true size effect.

(3) **No industry classification:** Industry membership—a standard control in asset pricing tests—was unavailable. Industry-neutral rankings could not be tested.

(4) **Non-redistributable data:** The RQAlpha market-data bundle is proprietary and not included in the project repository. This limits end-to-end reproducibility from raw data. The `data/evidence.json` file serves as the authoritative publication layer, providing all validated results with full provenance.

(5) **Sample consumption:** The 2022–2024 period was used as the third validation fold. A genuinely unseen forward test would require data from July 2026 onward. The consumed sample cannot be reused for further hypothesis testing without inflating Type I error rates.

(6) **Cost model:** Flat per-trade cost assumptions were used. In practice, costs vary with trade size, market conditions, and stock liquidity. The 31bp base scenario likely underestimates true costs in the lowest-liquidity quintile, meaning reported Q1 returns are upper bounds.

(7) **Model family:** Only LightGBM was tested. Other model classes (neural networks, random forests, linear models with interaction terms) might extract different information from the same features. However, given the feature family's fundamental limitation (OHLCV only), a model-class change was not expected to address the core problem.

(8) **Factor breadth:** Only 20 technical factors were used. Factor discovery algorithms, alternative technical indicators, or order-book features were not explored. However, the project's conclusion is about the information content of OHLCV data generally, not about a specific factor set.

(9) **No capacity model:** A formal market-impact model was not constructed. Capacity estimates are based on ADV20 quintile boundaries and should be treated as approximate.

(10) **Survivorship treatment:** The PIT universe included delisted stocks, but delisting returns were not explicitly modeled. Stocks that delisted during a holding period were treated as total losses, which is conservative.

11 Conclusion

Daily OHLCV features contain genuine cross-sectional ranking information. The frozen baseline retained a mean out-of-sample IC of 0.108 with Newey–West $t = 10.5$. Within the lowest-liquidity quintile, the model demonstrated real stock-selection ability (+2.7 monthly excess).

However, this ranking information did not survive the transition to executable, cost-aware, liquidity-filtered portfolios. In the liquid U50 and U30 universes, the strategy produced negative absolute returns (-0.9 net annual). Longer prediction horizons did not solve the concentration problem.

The project's contribution is not a trading signal. It is a demonstration that clean negative results are more valuable than profitable-looking lies; that statistical predictability and economic investability are fundamentally different properties; and that rejecting a favored hypothesis is the correct outcome of honest inquiry.

The most valuable output of this project was not a trading signal. It was a research system capable of proving me wrong.

A Evidence Map

Key evidence metrics:

- `target_leak_before_ic`: IC 0.940 → `EXP_TARGET_SHIFT_01` → `results/target_shift_audit.csv`
- `target_leak_after_ic`: IC 0.100 → `EXP_TARGET_SHIFT_01` → `results/target_shift_audit.csv`
- `frozen_baseline_mean_ic`: IC 0.108 → `EXP_BASELINE_V43` → `results/baseline_v43_summary.csv`
- `u100_raw_net_annual`: +6.8 → `EXP_DP1_PORTFOLIO` → `results/dp1_portfolio_summary.csv`
- `u50_resid_net_annual`: -0.9 → `EXP_STEP5_ABLATION` → `results/step5_comparison.csv`
- `q1_within_excess`: +2.7 → `EXP_DP1A_QUINTILE` → `results/dp1a_quintile.csv`
- `iid_null_mean_ic`: -0.005 → `EXP_NULL_IID_01` → `results/stage1_null_results.csv`
- `h3_u50_net_annual`: -21.7 → `EXP_MH1R_HORIZON` → `results/mh1r.csv`

B Experiment Ledger

1. **Target Timing**: IC 0.940 → 0.100. `EXP_TARGET_SHIFT_01`. **Invalid → Validated.**
2. **Cross-Fit Audit**: IC 0.760 → 0.120. `EXP_CROSSFIT_01`. **Invalid → Validated.**
3. **Frozen Baseline**: Mean OOS IC 0.108, NW $t = 10.5$. `EXP_BASELINE_V43`. **Validated.**
4. **Null Infrastructure**: iid null IC ≈ 0, permutation correlation ≈ 0. `EXP_NULL_IID_01`. **Validated.**
5. **Portfolio Diagnostics**: U100 +6.8%, U50 -0.9%. `EXP_DP1_PORTFOLIO`. **Q1-driven.**
6. **Residual Ablation**: RESID improves marginally, cannot reach profitability. `EXP_STEP5_ABLATION`. **Rejected.**
7. **Multi-Horizon**: H3 -21.7%, H6 -15.2% U50 net. `EXP_MH1R_HORIZON`. **Rejected.**
8. **Termination**: OHLCV-only core strategy closed. **Recorded.**

C Reproducibility

C.1 What Is Reproducible

- **Evidence validation:** All metrics in this report can be validated by running `python scripts/validate_evi` and `python -m unittest discover -s tests` from the repository root.
- **Website rendering:** The interactive landing page (`index.html`) runs with `python scripts/serve.py` and loads all displayed values from `data/evidence.json`.
- **Report metrics:** All numerical claims in this report are generated from `data/evidence.json` by `report/build.py`. The generated macros in `report/../generated/` are the sole source of metric values.

C.2 What Is Not Fully Reproducible

- **Model training:** The original LightGBM training runs require the RQAlpha market-data bundle (~3.3GB), which is proprietary and not redistributable. Training scripts are included but data-dependent.
- **Raw data:** Daily OHLCV data is not included in the repository.
- **Prediction artifacts:** Per-stock, per-month prediction files are referenced by path in `evidence.json` but stored locally only.

C.3 Repository

`https://github.com/DresdenGman/the-backtest-that-lied`

Commit `70e7ec7` (v1.0.0) contains the frozen landing page and evidence package. This report corresponds to release v1.1.0.

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